Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

For this project use concepts such as dataclasses, lists, functions, and conditionals. I expect to use loops to display and organize assignments, if statements to check whether assignments are completed or overdue, and lists to store multiple assignments and study goals

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the project will be creating the assignment and study goal dataclasses and allowing users to enter information, such as assignment titles, subjects, and due dates.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think the tasks that will require the most time and effort are creating the different Drafter routes, managing the program state, and making sure the assignment and study goal lists update correctly.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions that might interfere with my work are my phone, social media, and other school assignments that may take up my time. I plan to minimize these distractions by putting my phone away, working in a quiet environment, and setting aside specific times to focus on the project.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

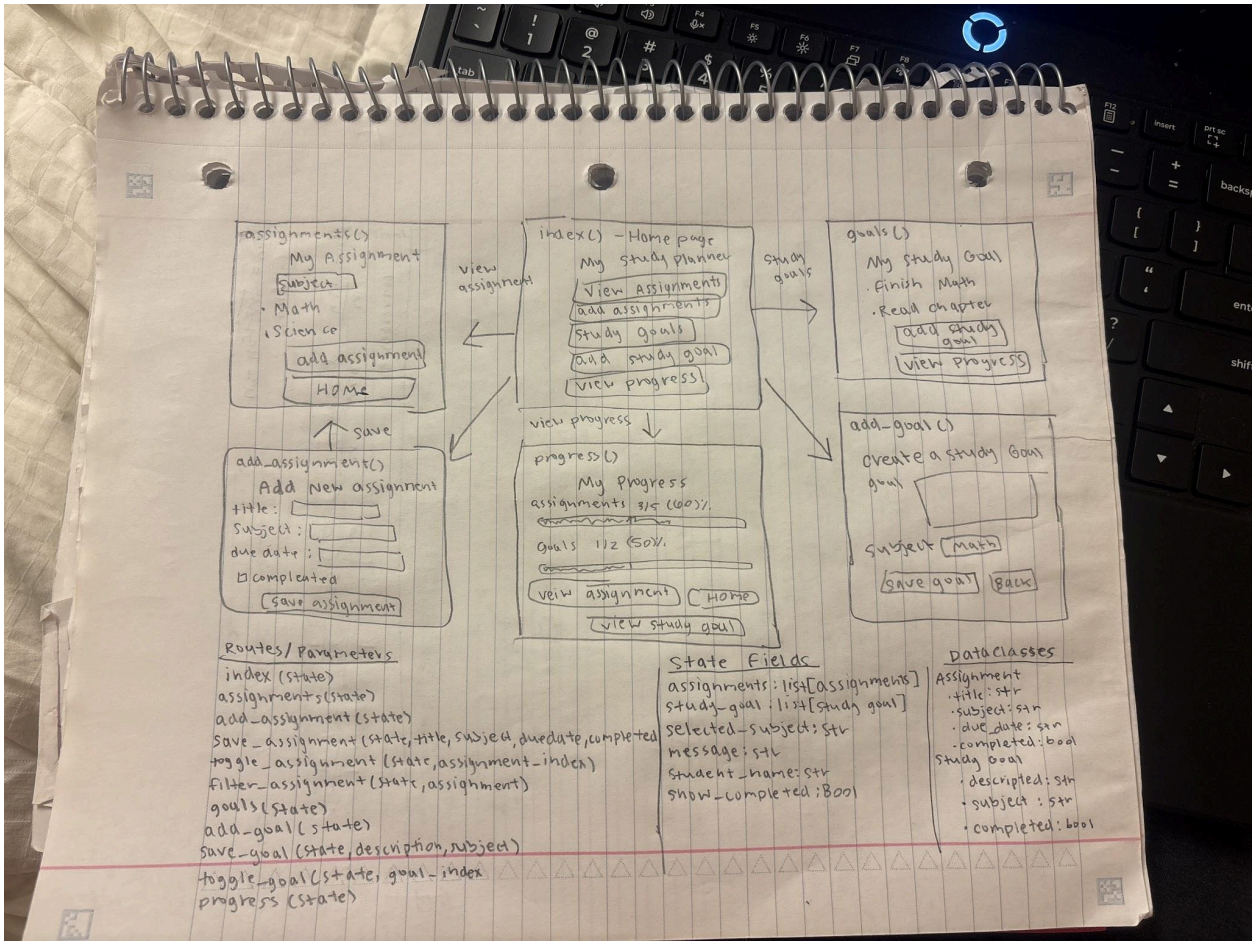
In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Add page

Search page

Display page


assignments()
My Assignment
Subject
• Math
• Science
add assignment
HOME
view assignment
index() - Home page
My study planner
View Assignments
add assignments
Study goals
add study goal
view progress
study goals
goals()
My study Goal
• Finish Math
• Read chapter
add study goal
view progress
save
add_assignment()
Add New assignment
title:
Subject:
due date:
☐ completed
save assignment
view progress
progress()
My Progress
assignments 3/4 (60%)
goals 1/2 (50%)
view assignment
HOME
view study goal
add_goal()
create a study Goal
goal
subject Math
save goal
Back
Routes/Parameters
index(state)
assignments(state)
add_assignment(state)
save_assignment(state, title, subject, duedate, completed)
toggle_assignment(state, assignment_index)
filter_assignment(state, assignment)
goals(state)
add_goal(state)
save_goal(state, description, subject)
toggle_goal(state, goal_index
progress(state)
State Fields
assignments: list[assignments]
study_goal: list[study goal]
selected_subject: str
message: str
student_name: str
show_completed: Bool
Dataclasses
Assignment
• title: str
• subject: str
• due_date: str
• completed: bool
Study Goal
• description: str
• subject: str
• completed: bool

++LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.